UET.CS2043 - Lập trình nâng cao

Refactoring

A case study

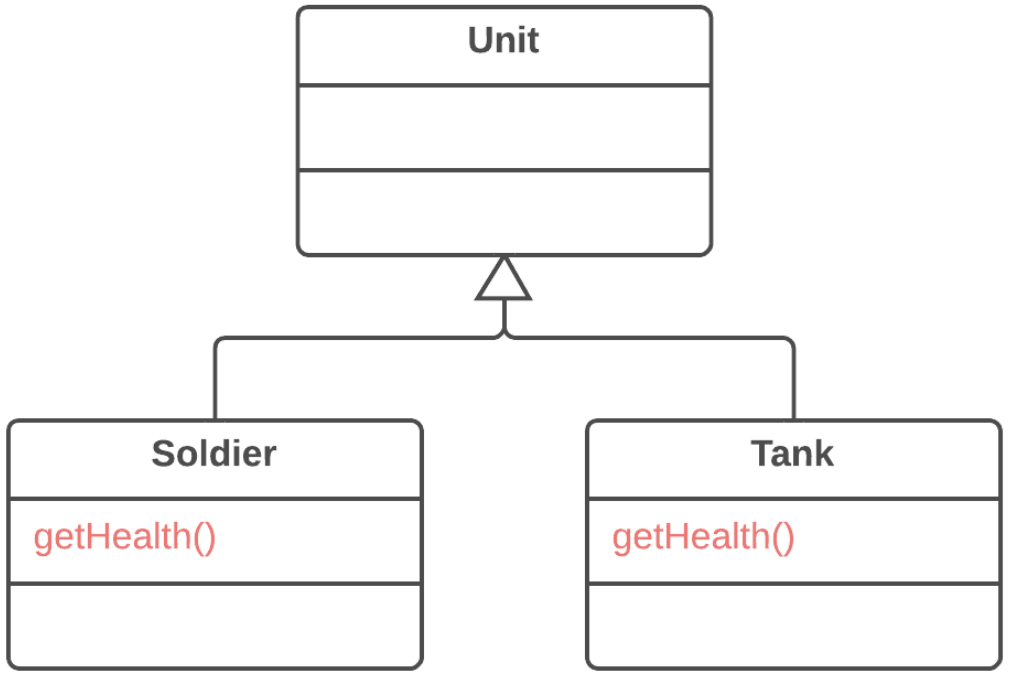
- John reviews a tank game project, examining its architecture and source code. He observes that the **high-level class design** is poorly structured and difficult to maintain.

A case study

Problems

- Some subclasses implement the **same** behavior, but this behavior is **not defined** in the base class.
- Methods defined in the base class are **not suitable** for all subclasses and are therefore frequently overridden.
- The base class defines a method, but subclasses **reimplement** the same behavior using methods with different names.

...



A case study

- John proposed that project management review and clean up the codebase, but the manager was not enthusiastic:

"The project is under heavy schedule pressure. We cannot clean up the code right now. Maybe in a few months."

- John also explained the situation to the developers. The programmers were very interested and clearly recognized the issues.

A case study

- The developers spent a few days refactoring the codebase.
  - They eliminated nearly half of the code, making it significantly more readable.
  - The team was very satisfied with the outcome and found that adding new classes and using existing ones became much faster and easier.

A case study

- Project manager was not satisfied
  - Before refactoring, the code was still functioning correctly.
  - The schedule was tight, and there was a significant amount of work to be done.
  - From management's perspective, the software needed to deliver the features requested by the customer, not clean or elegant code.
- The developers had spent two days without contributing to new features that the system was expected to deliver in the coming months.

A case study

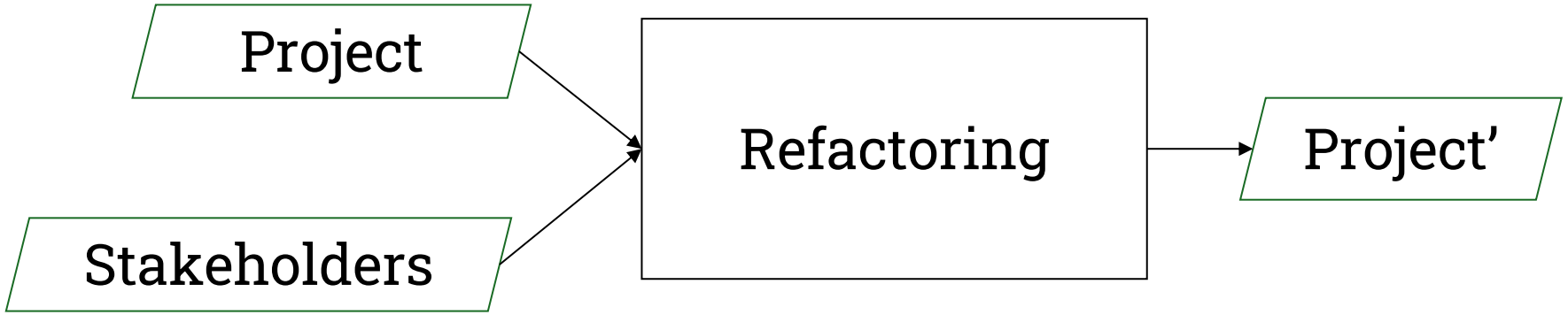
- Six months later, the project failed, largely because the codebase had become too complex to debug or adapt to achieve acceptable performance.

Code smell

- Code smells are usually not bugs
- Identify design deficiencies that negatively impact development speed and elevate the risk of bugs or system failures
- Kind of code smells
  - Condition & Parameter
  - Method
  - Field
  - Class/interface …

Refactoring

- Refactoring (noun): a change made to the internal structure of software to make it easier to understand and cheaper to modify without changing its observable behavior



Condition & Parameter

Decompose Conditional

- The longer a piece of code is, the harder it's to understand.

```
if (date.Before(SUMMER_START) || date.After(SUMMER_END))
  charge = quantity * _winterRate + _winterServiceCharge;
else
  charge = quantity * _summerRate;
```

```
if (NotSummer(date))
  charge = WinterCharge(quantity);
else
  charge = SummerCharge(quantity);
```

Long parameter list

```
public void SaveHomeAddress(string name,
                            string homeAddress,
                            string country,
                            string email,
                            string fileLocation)
{
}
```

```
public void SaveHomeAddress(AddressDetails addressDetails)
```

Method refactoring

Extract Method

- **Problem**: You have a code fragment that can be grouped together
- **Why Refactor**: The more lines found in a method, the harder it's to figure out what the method does
- **Solution**:
  - Move this code to a separate new method (or function)
  - Replace the old code with a call to the method.

```
public void PrintClient(Client client) {
    PrintBanner();
    // print client details
    Console.Write("Name: " + client.Name);
    Console.Write("Address: " + client.Address);
}
```

```
public void PrintClient(Client client) {
    PrintBanner();
    PrintClientDetails(client);
}

private void PrintClientDetails(Client client) {
    Console.Write("Name: " + client.Name);
    Console.Write("Address: " + client.Address);
}
```

1

2

Inline Method

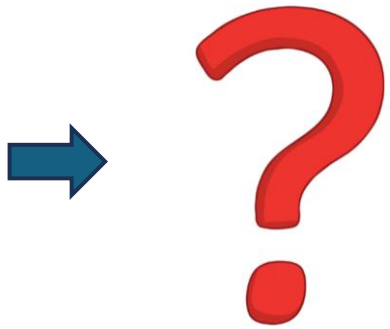
- Problem: a method body is more obvious than the method itself
- Solution: Replace calls to the method with the method's content and delete the method itself
- Benefit: By minimizing the number of unneeded methods, you make the code more straightforward.

```
class PizzaDelivery {
    1 usage
    int numberOfLateDeliveries;
    int getRating() {
        return moreThanFiveLateDeliveries() ? 2 : 1;
    }
    1 usage
    boolean moreThanFiveLateDeliveries() {
        return numberOfLateDeliveries > 5;
    }
```

```
class PizzaDelivery {
    1 usage
    int numberOfLateDeliveries;
    int getRating() {
        return numberOfLateDeliveries > 5 ? 2 : 1;
    }
//    boolean moreThanFiveLateDeliveries() {
//        return numberOfLateDeliveries > 5;
//    }
}
```

Inline Method - Exercise

```
abstract class Shape
{
  public void Draw()
  {
    try { /*draw*/ }
    catch (Exception e)
    {
      LogError(e);
    }
  }
  private static void LogError(Exception e)
  {
    File.WriteAllText(@"c:\Errors\Exception.txt", e.ToString());
  }
}
```



Inline Method - Exercise

```
abstract class Shape
{
  public void Draw()
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    try { /*draw*/ }
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  {
    File.WriteAllText(@"c:\Errors\Exception.txt", e.ToString());
  }
}
```

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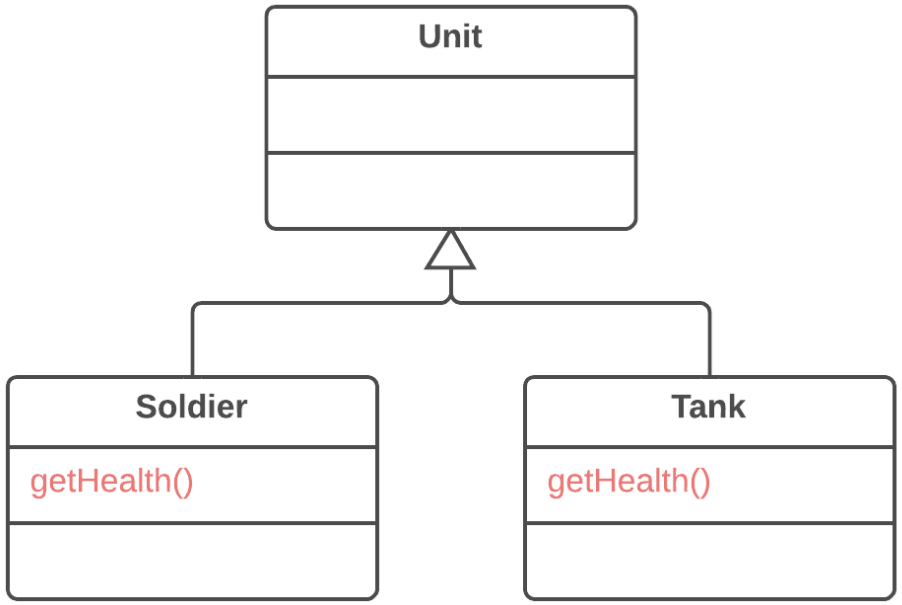
abstract class Shape
{
  public void Draw()
  {
    try { /*draw*/ }
    catch (Exception e)
    {
      File.WriteAllText(@"c:\Errors\Exception.txt", e.ToString());
    }
  }
}
```

Rename method

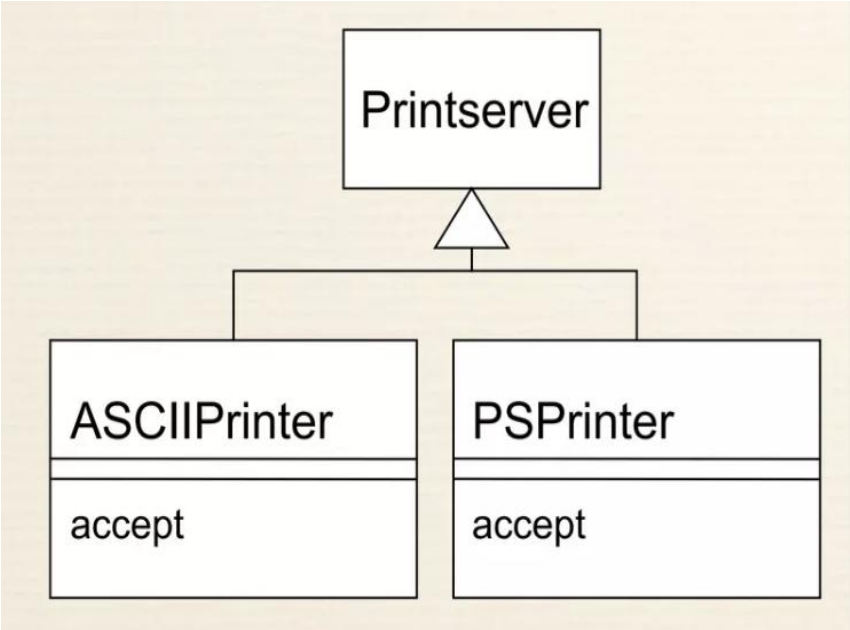
- Method name does not show its intention

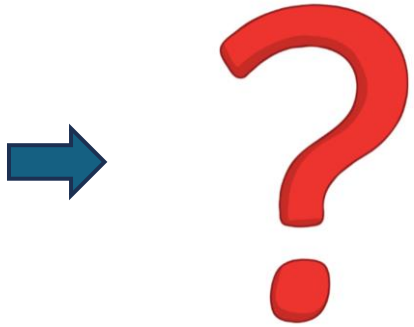
Pull up method

- **Problem**: Your subclasses have methods that perform similar work
- **Solution**: Make the methods identical and then move them to the relevant superclass
- **Why Refactor**: Subclasses grew and developed independently of one another, causing identical (or nearly identical) fields and methods.

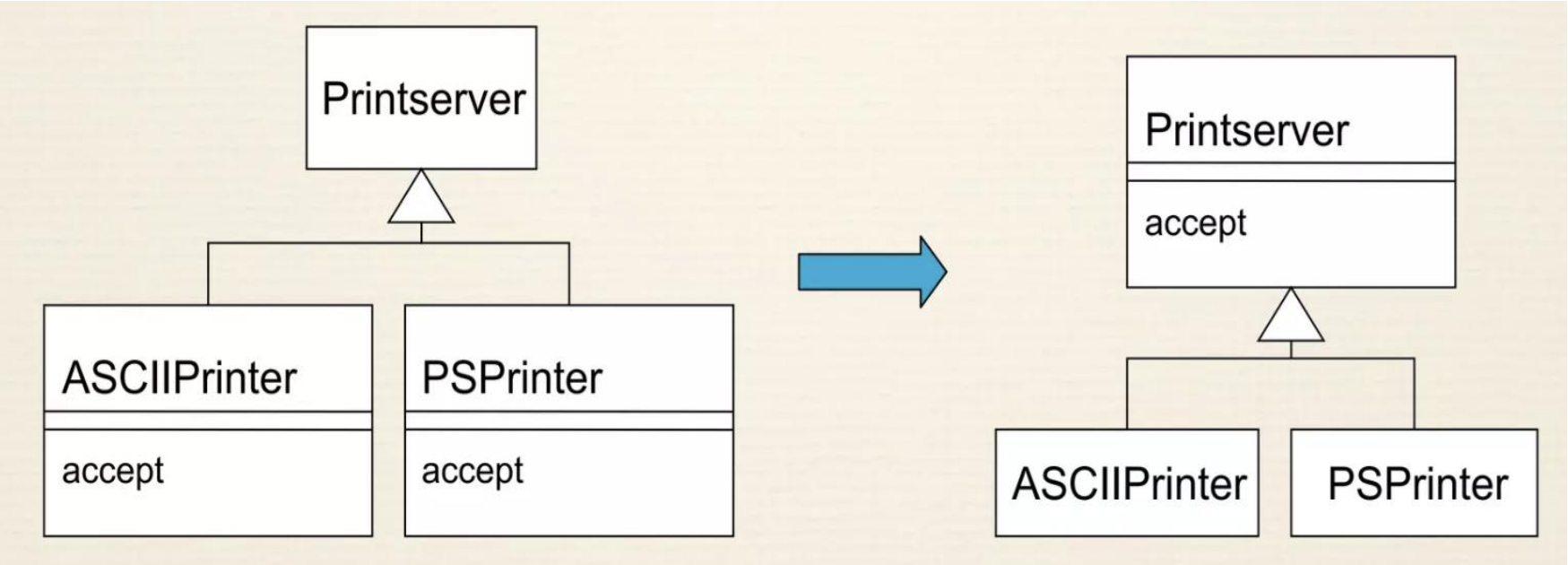


Pull up method – Exercise





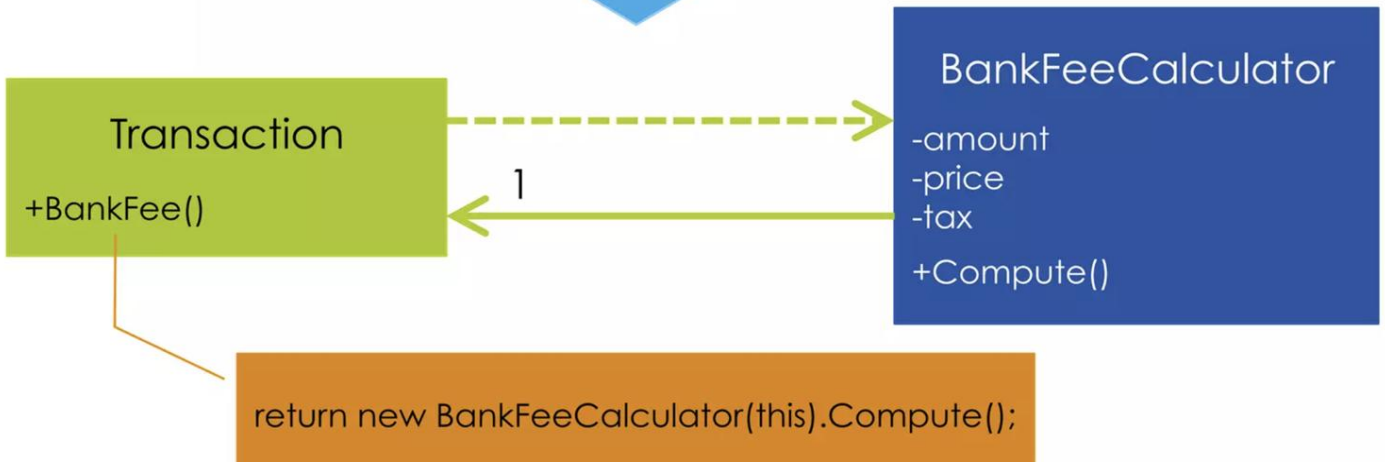
Pull up method – Exercise



Replace method with method objects

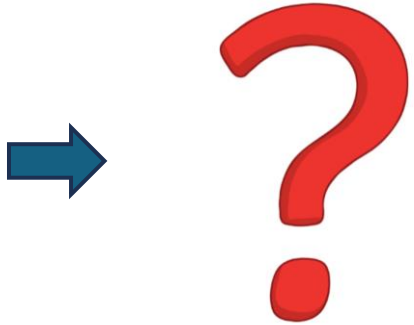
- **Problem**: You have a **long method**
- **Solution**
  - Transform the method into a separate class so that the local variables become fields of the class
  - Then you can split the method into several methods within the same class
- **Drawback**: Another class is added, increasing the overall complexity of the program.

```
class Transaction {
    public double BankFee() {
        double amount;
        double price;
        double tax;
        // long computations
    }
```



Replace method with method objects - Exercise

```java
class Order {
  // ...
  public double price() {
    double primaryBasePrice;
    double secondaryBasePrice;
    double tertiaryBasePrice;
    // Perform long computation.
  }
}
```



Replace method with method objects - Exercise

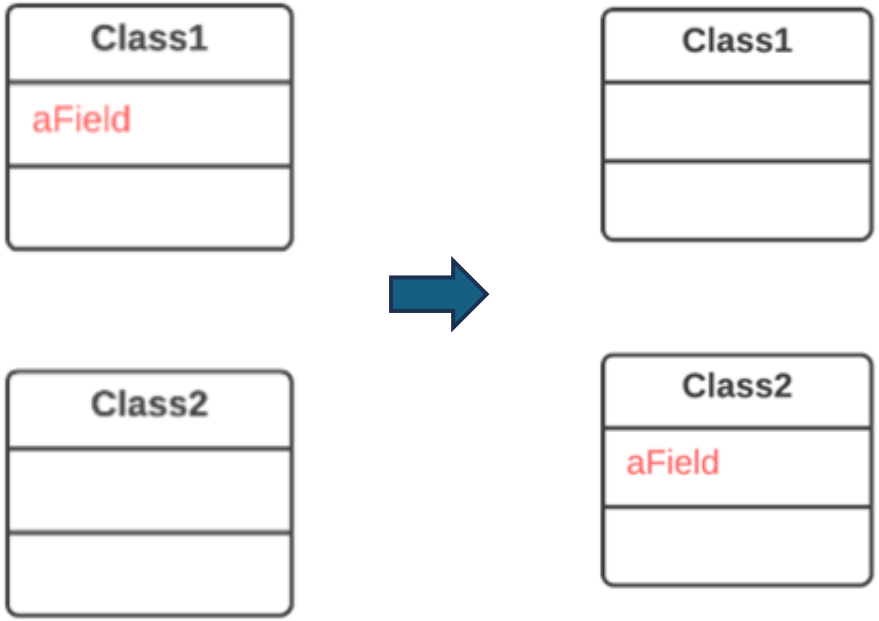
```java
class Order {
  // ...
  public double price() {
    double primaryBasePrice;
    double secondaryBasePrice;
    double tertiaryBasePrice;
    // Perform long computation.
  }
}
```

```java
class Order {
  // ...
  public double price() {
    return new PriceCalculator(this).compute();
  }
}

class PriceCalculator {
  private double primaryBasePrice;
  private double secondaryBasePrice;
  private double tertiaryBasePrice;

  public PriceCalculator(Order order) {
    // Copy relevant information from the
    // order object.
  }

  public double compute() {
    // Perform long computation.
  }
}
```

Field refactoring

Move Field

- **Problem**: A field is used more in another class than in its own class
- **Solution**: Create a field in a new class and redirect all users of the old field to it



Replace magic number with constant

- **Problem**: Your code uses a number that has a certain meaning to it
- **Solution**: Replace this number with a constant that has a human-readable name explaining the meaning of the number

```
double PotentialEnergy(double mass, double height) {
  return mass * 9.81 * height;
}
```

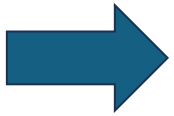
```
static const double GRAVITATIONAL_CONSTANT = 9.81;
double PotentialEnergy(double mass, double height) {
  return mass * GRAVITATIONAL_CONSTANT * height;
}
```

Encapsulate fields

- Create and use getter/setter to access field

```java
//File Class.java
public class Class {
    public String aString;
}
```

```java
//File AnotherClass.java
public class AnotherClass {
    public Class aClass;
    public void method() {
        aClass.aString="string";
    }
}
```



```java
//File Class.java
public class Class {
    private String aString;
    public void setaString(String aString) {
        this.aString = aString;
    }
    public String getaString() {
        return aString;
    }
}
```

```java
//File AnotherClass.java
public class AnotherClass {
    public Class aClass;
    public void method() {
        aClass.setaString("string");
    }
}
```

Encapsulate fields - Exercise

```java
public class NormalFieldClass {
    public String name;

    public static void main(String[] args)
    {
        NormalFieldClass example1 = new NormalFieldClass();
        example1.name = "myName";
        System.out.println("My name is " + example1.name);
    }
}
```

```java
public class EncapsulatedFieldClass {
    private String name;

    public String getName()
    {
        return name;
    }

    public void setName(String newName)
    {
        name = newName;
    }

    public static void main(String[] args)
    {
        EncapsulatedFieldClass example1 = new EncapsulatedFieldClass();
        example1.setName("myName");
        System.out.println("My name is " + example1.getName());
    }
}
```

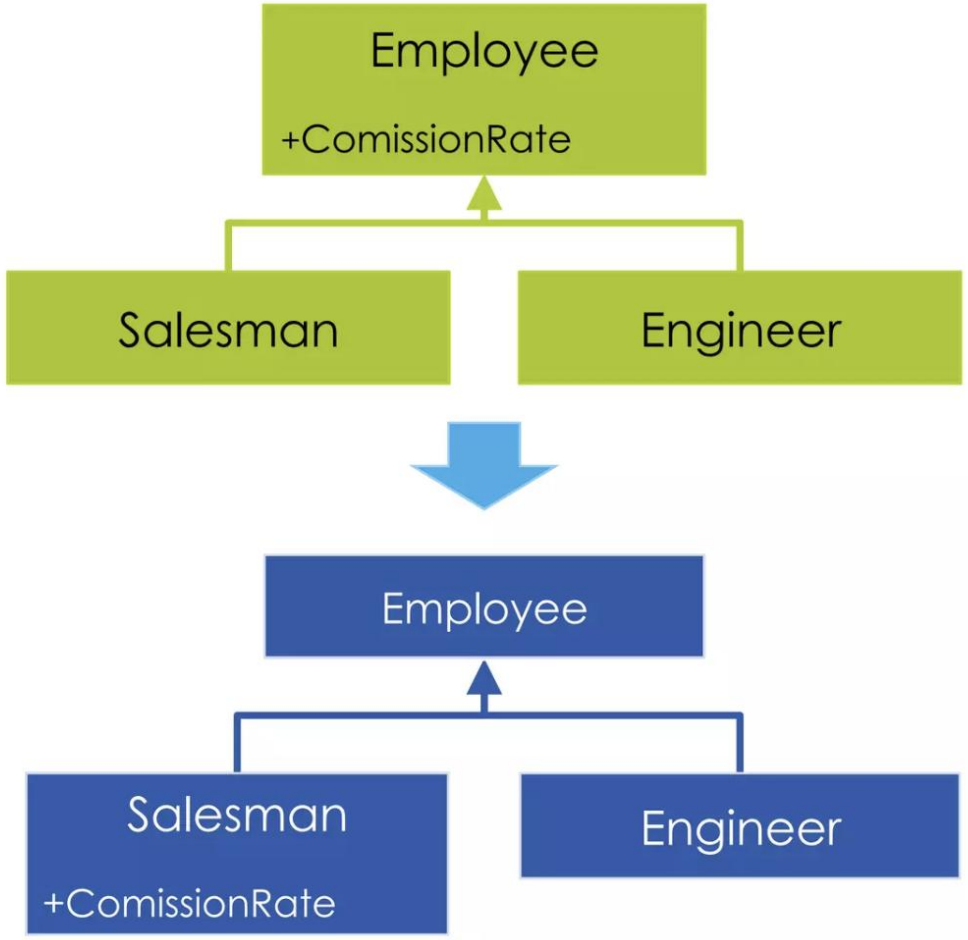
Explaining Variable

```
if ((platform.toUpperCase().indexOf("MAC") > -1) &&
    (browser.toUpperCase().indexOf("IE") > -1) &&
    isInitialized() && resize > 0) {
    // do something
}
```

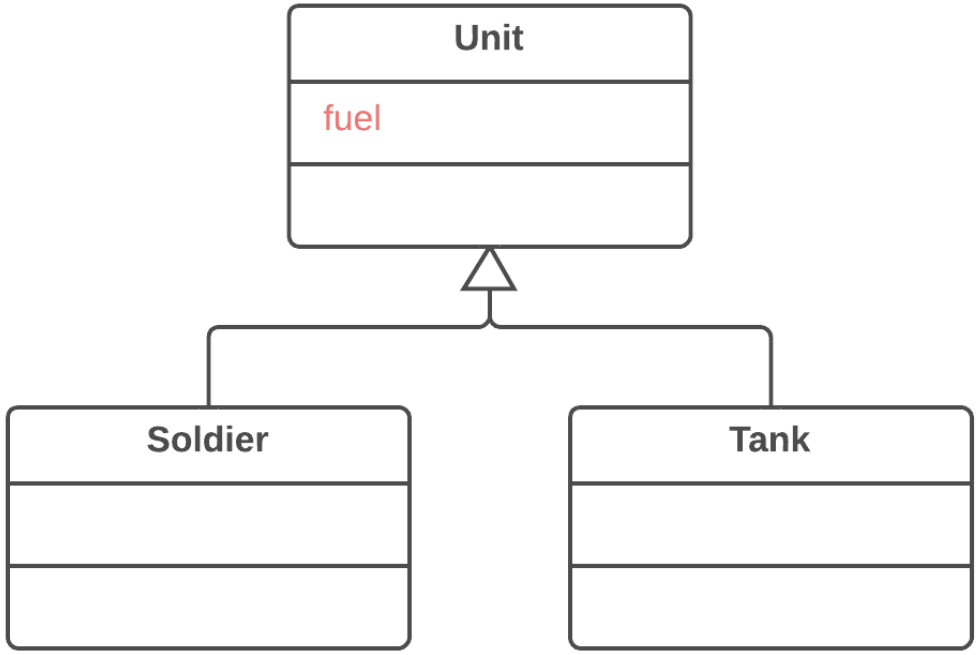
```
var isMac = platform.toUpperCase().indexOf("MAC") > -1;
var isIEBrowser = browser.toUpperCase().indexOf("IE") > -1;
var isResized = resize > 0;
if (isMac && isIEBrowser && isInitialized() && isResized){
    // do something
}
```

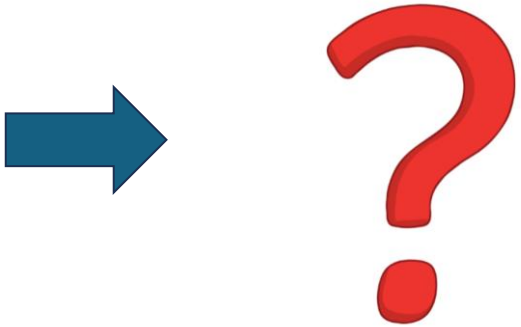
Push down field

- **Problem**: Is a field used only in a few subclasses?
- **Solution**: Move the field to these subclasses

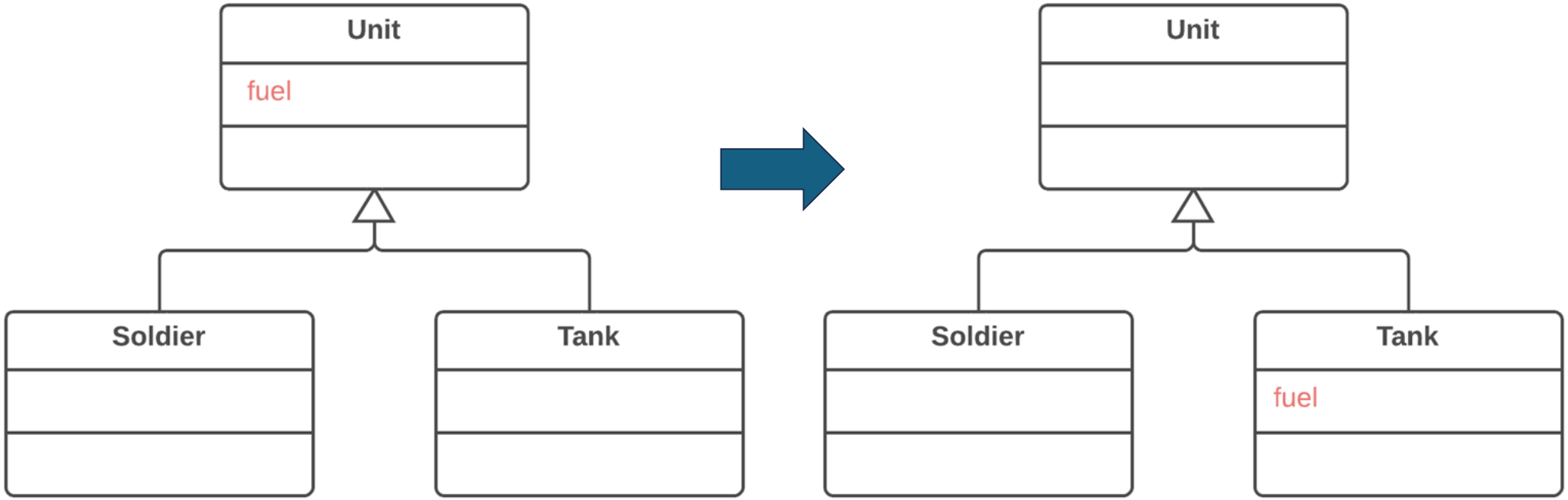


Push down field - Exercise





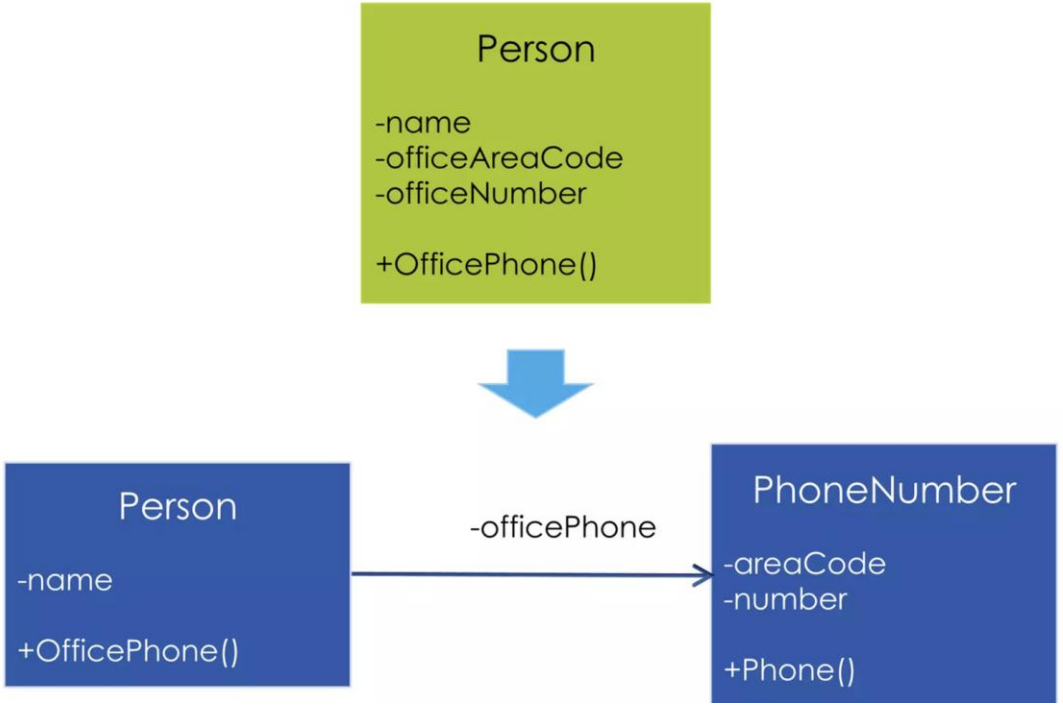
Push down field - Exercise



Class/interface refactoring

Extract Class

- **Problem**: When one class does the work of two, awkwardness results (violate Single Responsibility Principle)
- **Solution**: Instead, create a new class and place the fields and methods responsible for the relevant functionality in it

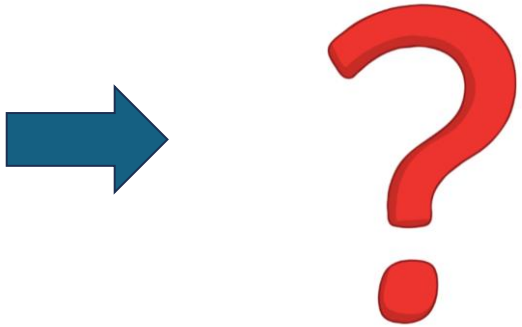


Replace conditional with polymorphism

- **Problem**: You have a conditional that performs various actions depending on object type or properties
- **Solution**: Create subclasses matching the branches of the conditional

Replace conditional with polymorphism

```
class Bird {
  // ...
  double getSpeed() {
    switch (type) {
      case EUROPEAN:
        return getBaseSpeed();
      case AFRICAN:
        return getBaseSpeed() - getLoadFactor() * numberOfCoc
      case NORWEGIAN_BLUE:
        return (isNailed) ? 0 : getBaseSpeed(voltage);
    }
    throw new RuntimeException("Should be unreachable");
  }
}
```



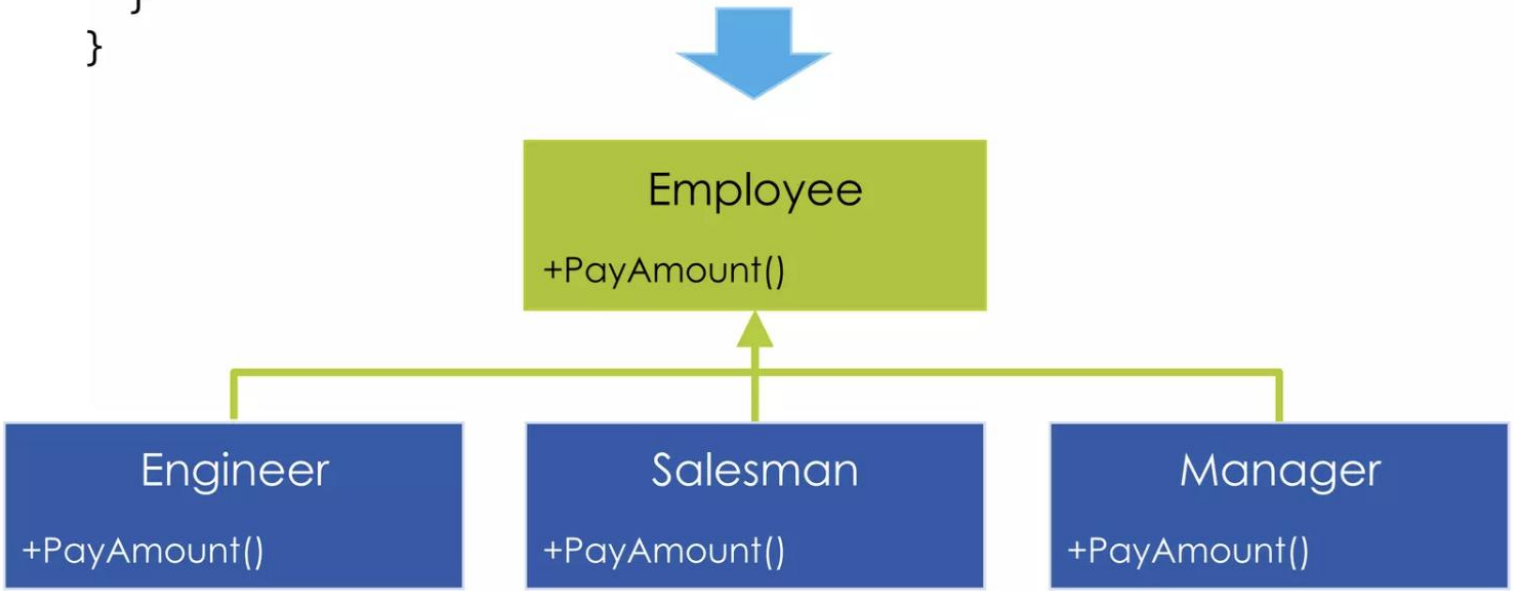
Replace conditional with polymorphism

```java
class Bird {
  // ...
  double getSpeed() {
    switch (type) {
      case EUROPEAN:
        return getBaseSpeed();
      case AFRICAN:
        return getBaseSpeed() - getLoadFactor() * numberOfCoc
      case NORWEGIAN_BLUE:
        return (isNailed) ? 0 : getBaseSpeed(voltage);
    }
    throw new RuntimeException("Should be unreachable");
  }
}
```

```java
abstract class Bird {
  // ...
  abstract double getSpeed();
}

class European extends Bird {
  double getSpeed() {
    return getBaseSpeed();
  }
}
class African extends Bird {
  double getSpeed() {
    return getBaseSpeed() - getLoadFactor() * numberOfCoconuts;
  }
}
class NorwegianBlue extends Bird {
  double getSpeed() {
    return (isNailed) ? 0 : getBaseSpeed(voltage);
  }
}

// Somewhere in client code
speed = bird.getSpeed();
```

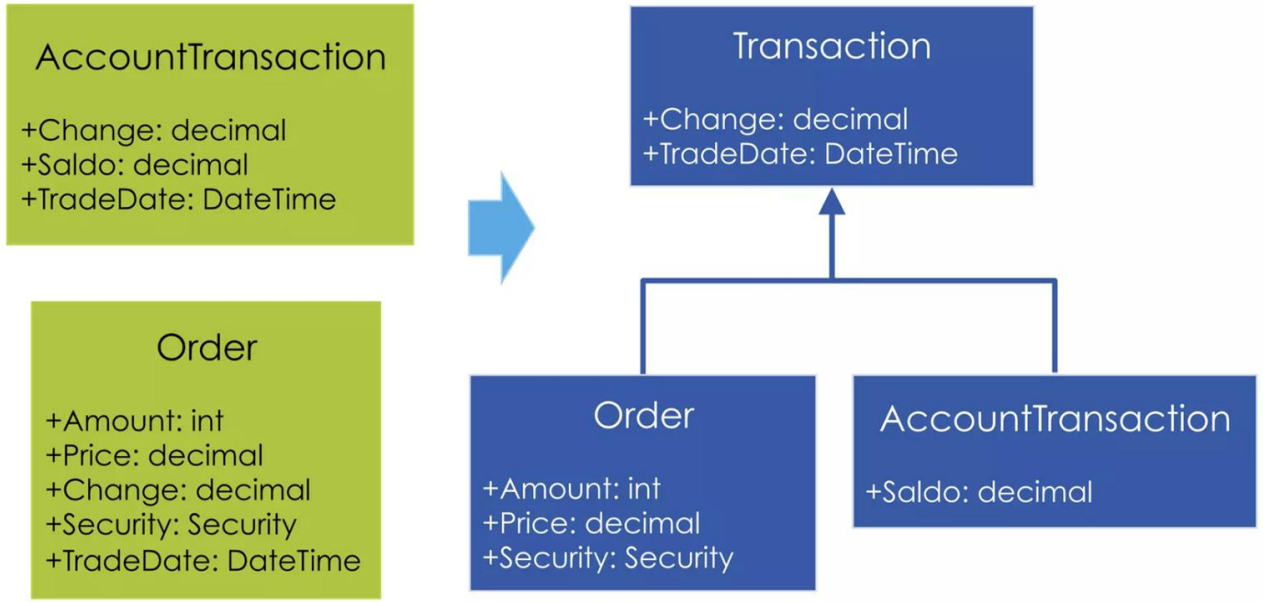
Replace conditional with polymorphism

```
class Employee {
  public double PayAmount() {
    switch (_type) {
      case EmployeeType.Engineer: return _salary;
      case EmployeeType.Salesman: return _salary + _commission;
      case EmployeeType.Manager: return _salary + _bonus;
      default: throw new WrongEmployeeTypeException();
    }
  }
```

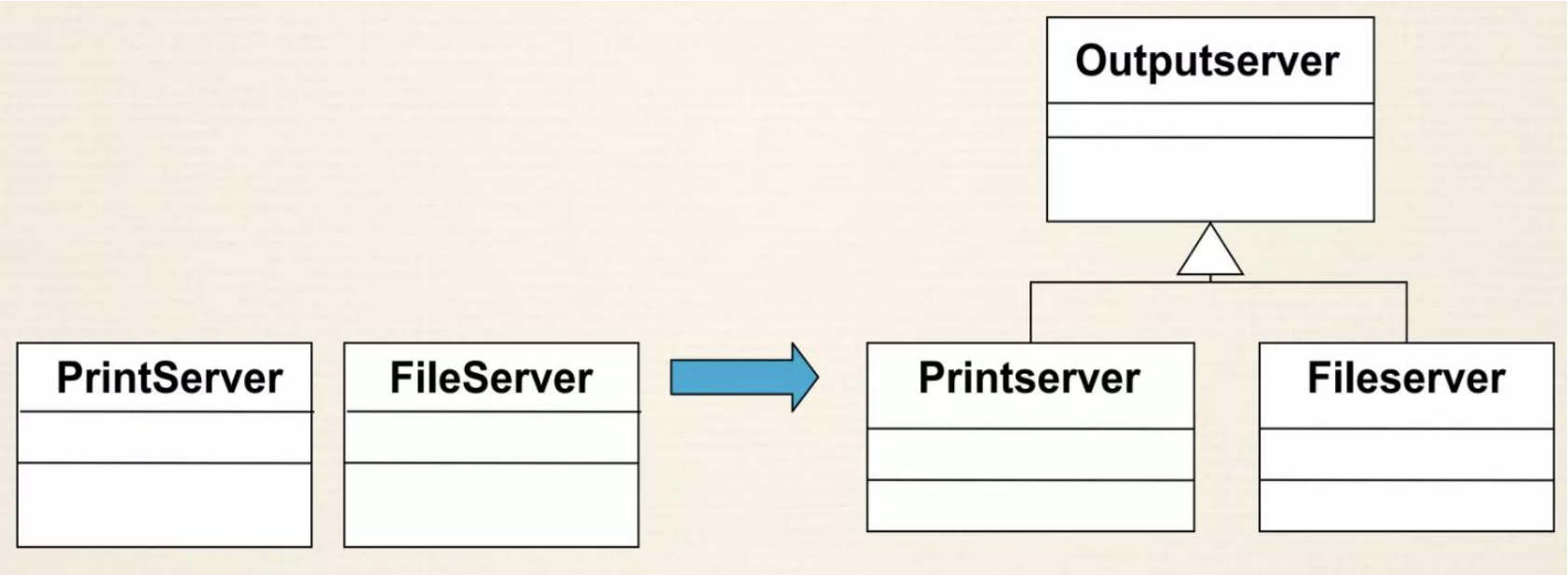


Extract superclass

- **Problem**: You have two classes with common fields and methods
- **Solution**: Create a shared superclass for them and move all the identical fields and methods to it

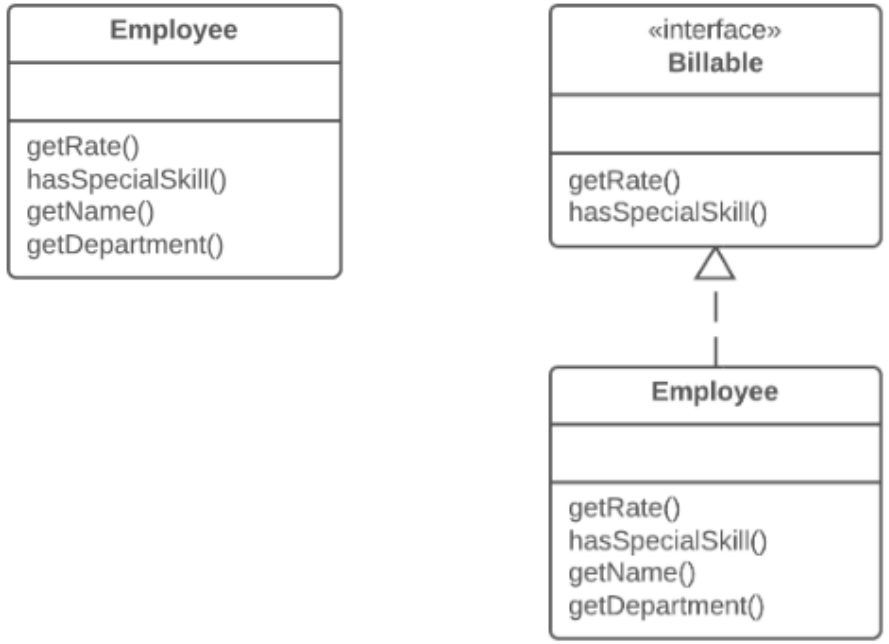


Extract superclass - Exercise



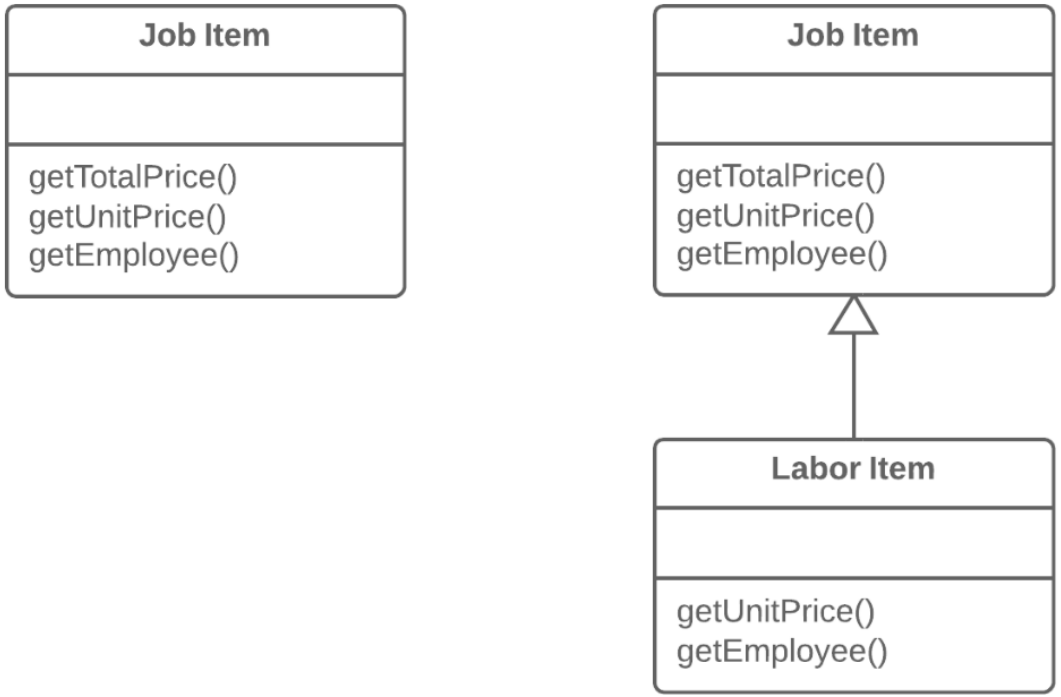
Extract interface

- **Problem**
  - Multiple clients are using the same part of a class interface
  - Another case: part of the interface in two classes is the same
- **Solution**
  - Move this identical portion to its own interface



Extract subclass

- **Problem**: A class has features that are used only in certain cases
- **Solution**: Create a subclass and use it in these cases

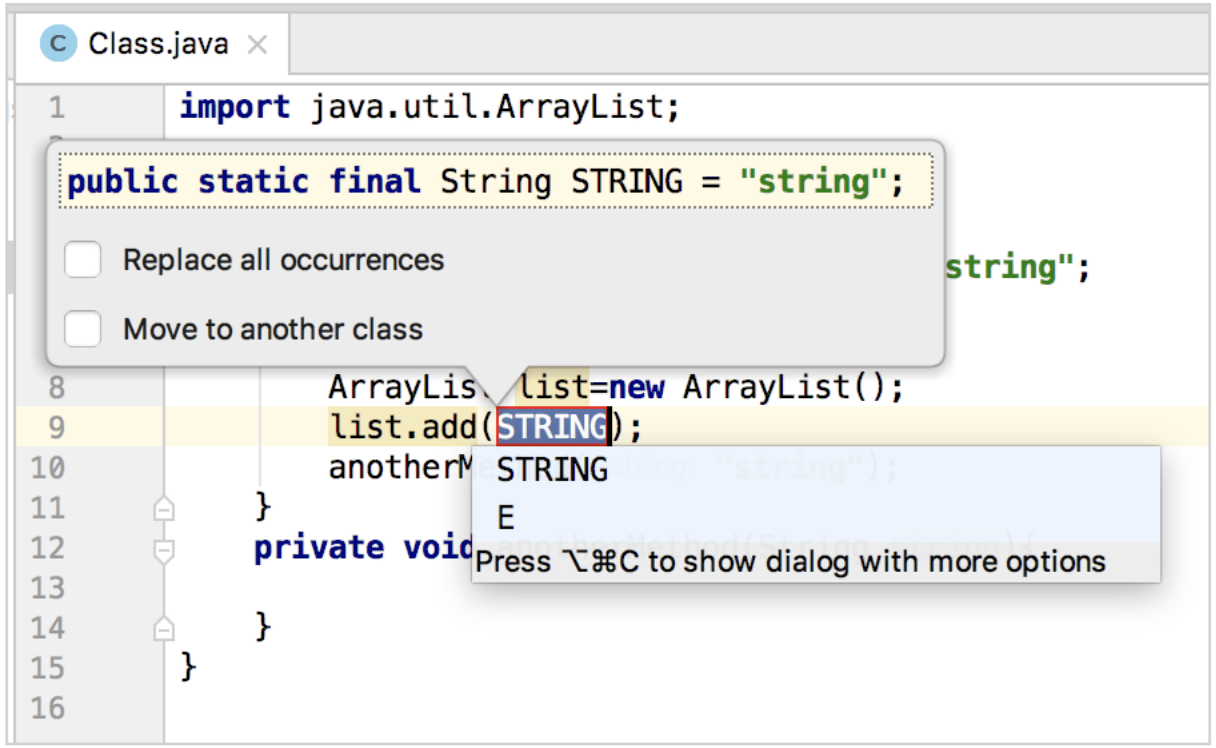


Tools

Intelj: https://www.jetbrains.com/help/idea/refactoring-source-code.html

Extract constant

```
public class Class {
    public void method() {
        ArrayList list = new ArrayList();
        list.add("string");
        anotherMethod("string");
    }

    private void anotherMethod(String string) {
    }
}
```




```
public class Class {
    private static final String STRING = "string";

    public void method() {
        ArrayList list = new ArrayList();
        list.add(STRING);
        anotherMethod(STRING);
    }

    private void anotherMethod(String string) {
    }
}
```

https://www.jetbrains.com/help/idea/extract-constant.html

Extract field

```java
public class Class {
    AnotherClass anotherClass;
    public void method() {
        int a = 1;

        int b = a + anotherClass.intValue();
        int c = b + anotherClass.intValue();
    }
}
```

```java
public class Class {
    public AnotherClass anotherClass;
    private int number;

    public void method() {
        int a = 1;
        number = anotherClass.intValue();
        int b = a + number;
        int c = b + number;
    }
}
```

References

1. “Refactoring: Improving the Design of Existing Code” (2nd edition), Martin Fowler, 2018
2. “Clean Code: A Handbook of Agile Software Craftsmanship”, Robert C. Martin, 2008
3. “Code Complete” (2nd Edition), Steve McConnell, 2004
4. “Industry experiences with large-scale refactoring”, James Ivers et al., 2022 (https://doi.org/10.1145/3540250.3558954)
5. https://refactoring.guru/refactoring
6. https://www.industriallogic.com/xp/refactoring/index.html

Any Questions